Mohammed Fazal Ur Rahman

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.40
M.S Education Academy	Class XII (Telangana State Board of Intermediate Education (TSBIE))	2024	97.40%
Mamatha High School, Nalgonda , Telangana	Class X (Board of Secondary Education, Telangana)	2022	9.70%

EXPERIENCE

AGC Asia Pacific (India) Pvt. Ltd. -- Computer Vision Research Intern

September 2026 - Present

PROJECTS

Research Paper RAG | GitHub

- Built a **Retrieval-Augmented Generation (RAG)** application for question answering over research papers using **LangChain**, **FAISS**, **BM25**, **HuggingFace embeddings (BGE)**, and **Google Gemini**.
- Implemented an end-to-end retrieval pipeline with **PDF loading, document chunking, vector indexing, hybrid retrieval (Dense + BM25)**, and **context-grounded response generation** with source-aware answers.

MiniGPT - Decoder-Only Transformer from Scratch | GitHub

- Built a **5.82M-parameter GPT-style decoder-only Transformer from scratch in PyTorch**, implementing causal multi-head self-attention, Pre-LayerNorm blocks, residual connections, learned positional embeddings, and GELU-based feed-forward networks.
- Trained a custom **Byte-Level BPE tokenizer (2,048 vocabulary)** and trained the model end-to-end on Tiny Shakespeare for autoregressive next-token prediction with a **128-token context window**.
- Implemented validation-based checkpointing, **temperature/top-k text generation**, and perplexity evaluation; achieved **4.004 validation loss and 54.82 perplexity**, with the best checkpoint selected at **3,250 training steps**.

Neural Machine Transliteration using Seq2Seq RNNs and Attention - GitHub | WandB

- Developed a character-level Neural Machine Transliteration system in **PyTorch** to convert Romanized words into native script using Seq2Seq Encoder-Decoder architectures.
- Implemented and compared **Vanilla Seq2Seq** and **Bahdanau Attention-based Seq2Seq models** with RNN, LSTM, and GRU cells, evaluating their impact on transliteration performance.
- Enhanced the attention model by incorporating PAD masking, teacher forcing, gradient clipping, and context vector concatenation for more stable training and improved sequence alignment.
- Achieved **73.99% character-level test accuracy** on the **Google Dakshina dataset** using the best-performing attention-based LSTM model.
- Built a complete evaluation and visualization pipeline, generating prediction tables, **attention heatmaps**, and training/validation performance curves to interpret model behavior.

Fashion-MNIST Classification using a Neural Network from Scratch - GitHub | WandB

- Implemented a **fully configurable feedforward neural network from scratch** using **NumPy** for Fashion-MNIST classification, achieving **88.6% test accuracy** without relying on deep learning frameworks such as PyTorch or TensorFlow.
- Developed **fully vectorized implementations** of forward propagation, backpropagation, mini-batch gradient descent, and gradient clipping to improve computational efficiency and training stability.

- Implemented and compared multiple optimization algorithms, including SGD, Momentum, Nesterov Accelerated Gradient (NAG), RMSProp, and Adam, along with Xavier and He weight initialization techniques.
- **Designed modular** and **flexible neural network architecture** supporting **variable hidden layers** and **neuron configurations**, and performed **hyperparameter tuning** and **experiment tracking** using **Weights & Biases (W&B) Sweeps**.

SKILLS

Languages - Python, C, C++

Frameworks - PyTorch, LangChain, NumPy, Pandas, Matplotlib, Scikit-learn

Databases - SQL, MySQL

Technologies - Git/GitHub, FAISS, Google Gemini API, WandB

Domains - Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Large Language Models(LLMs), Retrieval-Augmented Generation (RAG), Data Structures & Algorithms, Object Oriented Programming,

CERTIFICATIONS

Convolutional Neural Networks By Andrew N.g

Structuring Machine Learning Projects By Andrew N.g

Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization By Andrew N.g

Neural Networks and Deep Learning By Andrew N.g

RELEVANT COURSES

Programming & Machine Learning Courses - Introduction to Programming, Artificial Intelligence, Data structures and applications, Algorithms and data structures lab.

Mathematics - Calculus, Elementary Linear algebra, Differential equations, Introduction to Probability, Introduction to Statistics

POSITIONS OF RESPONSIBILITY

Finance & Partnerships Coordinator | TEDxIITHyderabad

- Coordinated sponsorship operations for the 10th edition of TEDxIITHyderabad, managing sponsor outreach, communication, and execution of activities.

Internship Coordinator | Office of Career Services, IIT Hyderabad

- Coordinated and ensured the smooth execution of internship interviews for IIT Hyderabad students.